



SafeNode Product and Security Overview

What SafeNode is, who it serves, and the security guarantees it is designed to provide.

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Product: Safenode

Classification: Internal / customer-safe where noted

Executive summary

Safenode is a passkey-first security service for identity, recovery, encrypted secrets, and team access. It is not only a password vault: it is a control plane for trusted authentication, encrypted personal vaults, recovery workflows, and shared team vaults.

The system is designed so the backend stores encrypted vault payloads, wrapped-key material, account metadata, device state, and subscription state, but does not receive the master password, raw vault key, recovery-kit secret, or WebAuthn PRF output.

Important: Safenode is a product security design, not a certification. Customers must still evaluate their own device, browser, identity provider, and operational risks.

Product thesis

- **Identity:** passkeys and WebAuthn provide phishing-resistant authentication.
- **Recovery:** recovery kits, device re-approval, and successor workflows address account continuity without server-side key escrow.
- **Secrets:** personal and team vaults hold encrypted entries and encrypted attachments.

- **Control:** device limits, audit events, rate limits, SSO, billing entitlements, and security telemetry govern access.

Security boundary

Protected secret	Where it exists	Server receives
Master password	User input and process memory only	No
Raw vault key	Client memory during an unlocked session	No
Recovery-kit secret	User-controlled recovery material	No
PRF output	Authenticator/browser assertion result	No
Encrypted vault blob	Client storage and server sync	Ciphertext only
PRF wrapped key	Server-side credential record	Ciphertext plus non-secret salt

Supported surfaces

- Web application served through Cloudflare Pages.
- Fastify backend deployed on Railway with Prisma/Postgres.
- iOS and Android native shells with platform passkey support and mobile API origin handling.
- Tauri desktop foundation for macOS, Windows, and Linux with browser-to-app authentication handoff.
- Browser extension and downloadable release artifacts subject to separate release gates.

Customer responsibilities

- Use a trusted device and keep its operating system and browser updated.
- Register more than one passkey or retain a recovery kit in a safe offline location.
- Verify production domains and never approve an unexpected passkey prompt.
- For teams, define administrators, member roles, offboarding, and recovery ownership before storing critical secrets.

Current release gates

- Live PRF passkey unlock test on supported authenticators.
- Email verification and password-reset delivery through the verified Resend domain.
- Paddle monthly/annual checkout and signed webhook processing.
- Cloudflare Pages, Railway health, database schema sync, and native association files.

- CI frontend/backend tests and security checks.